
SSM Optimization

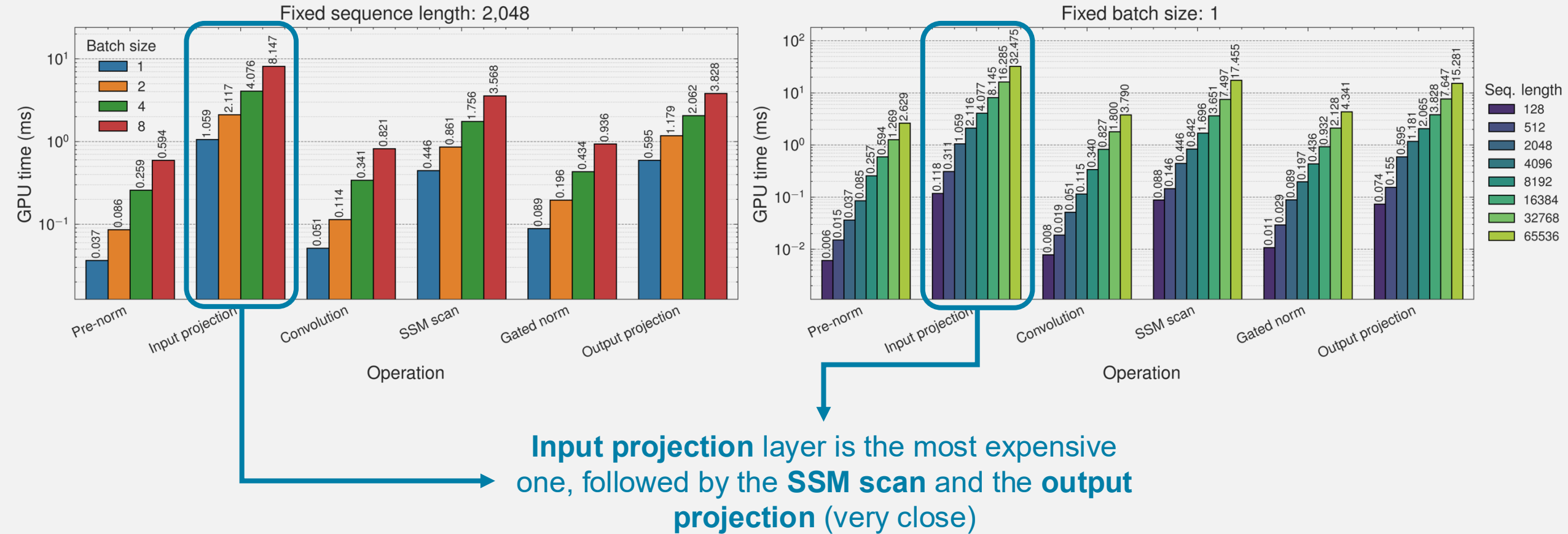
Weekly Meeting

31/07/26

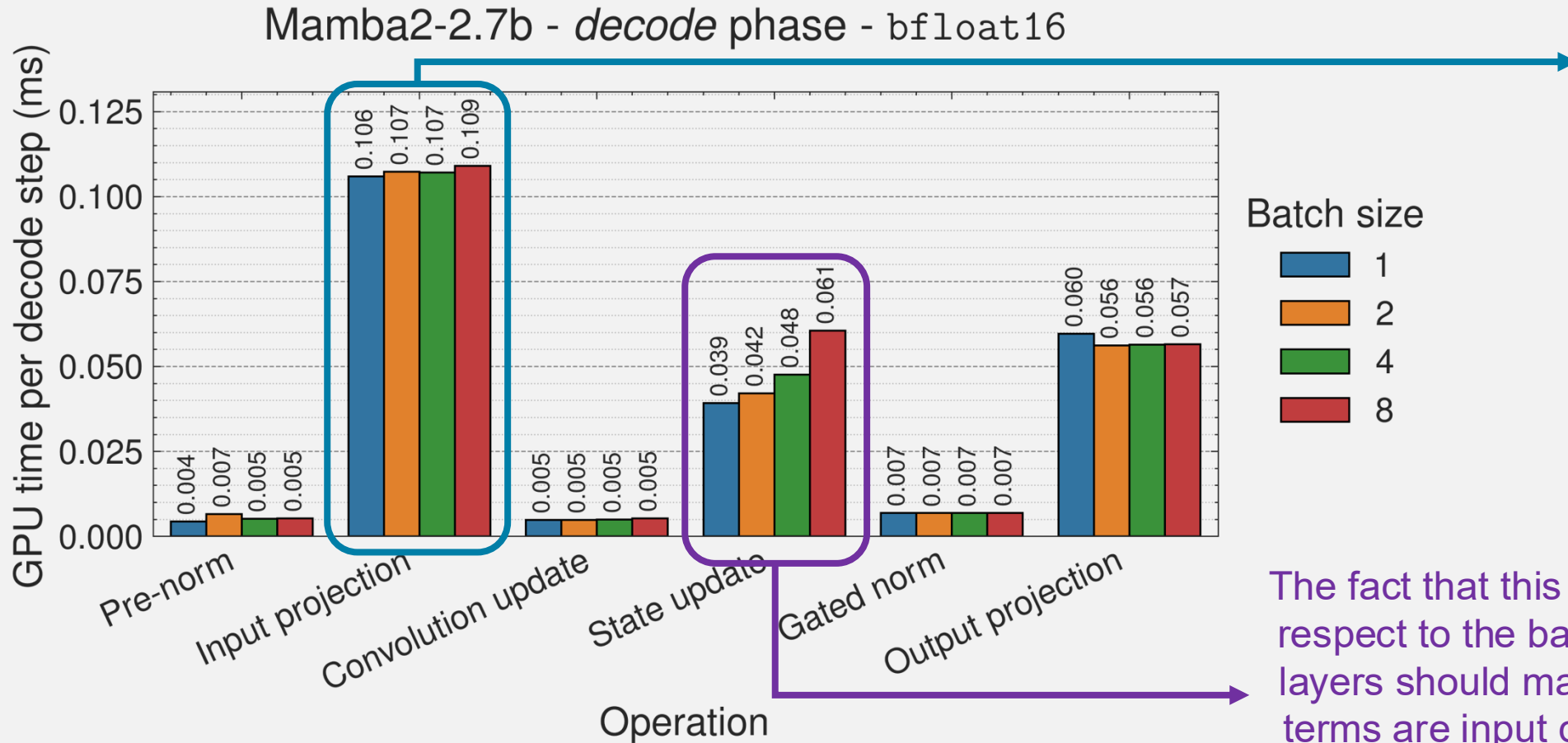
Profiling

Mamba-2-2.7b block / prefill

Mamba2-2.7b - prefill phase - bfloat16



Mamba-2-2.7b block / decode



Input projection layer remains the most expensive one, followed by the **output projection** and only then the state update

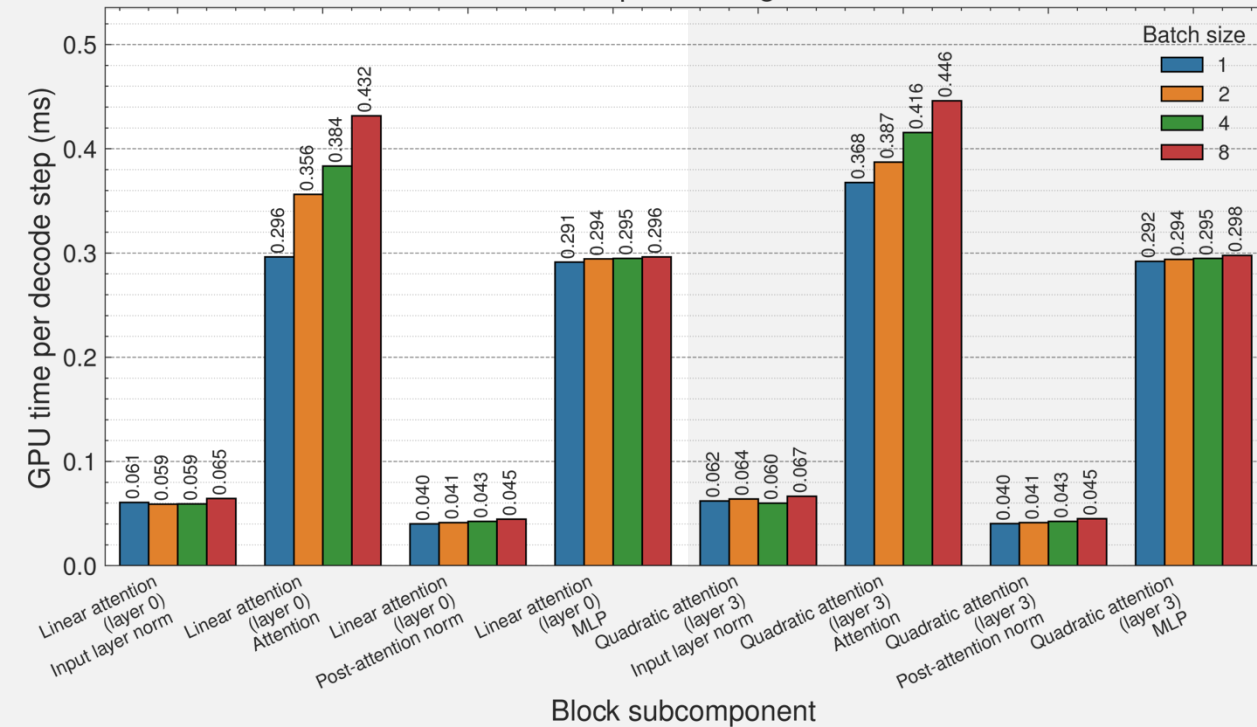
The fact that this cost is not constant with respect to the batch size as for the other layers should make sense, as the all the terms are input dependent, thus it is not possible to reuse weights

Profiling

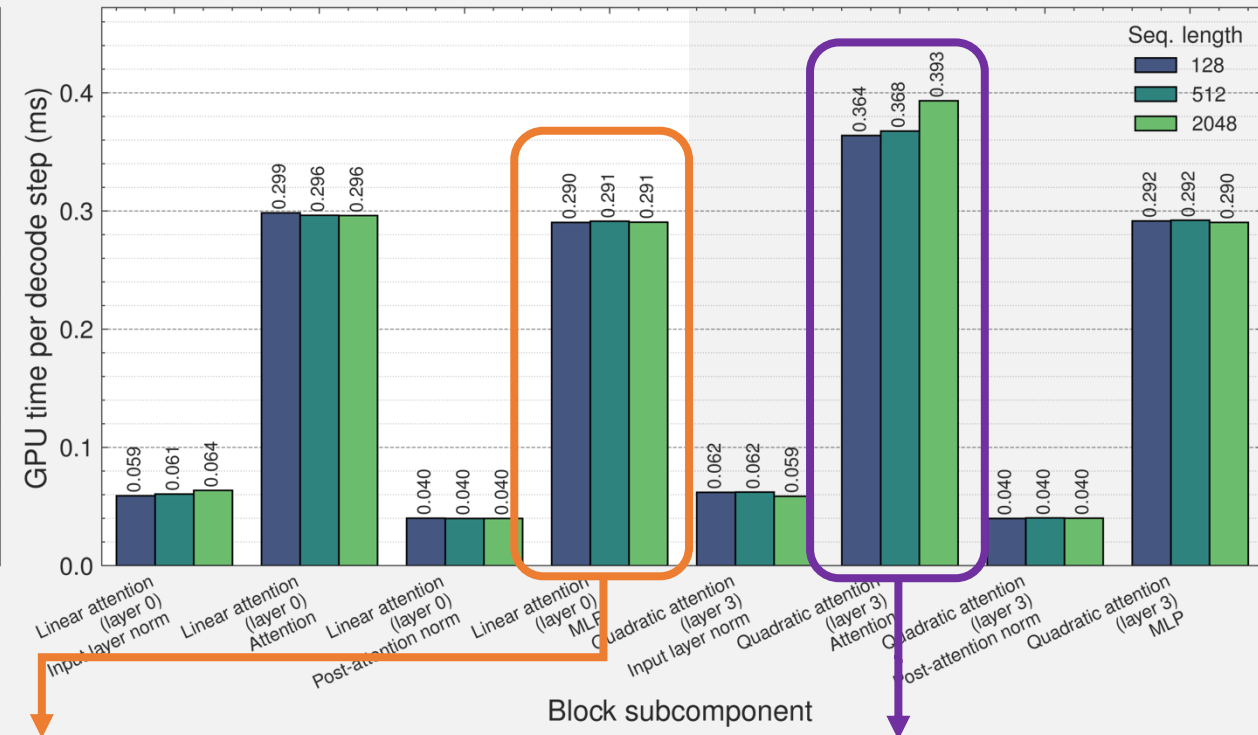
Qwen3.5-4B block / decode

Qwen3.5-4B - decode phase - bfloat16

Fixed sequence length: 512



Fixed batch size: 1



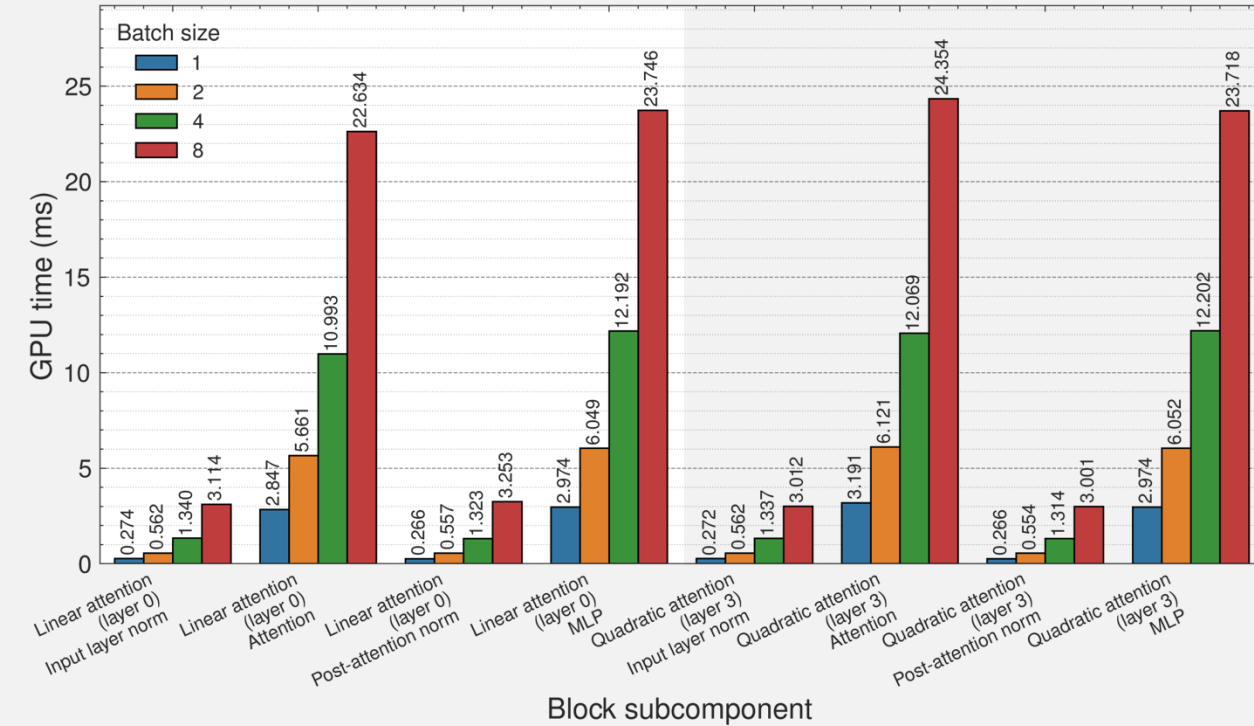
Constant with respect to
sequence length

Small increase...

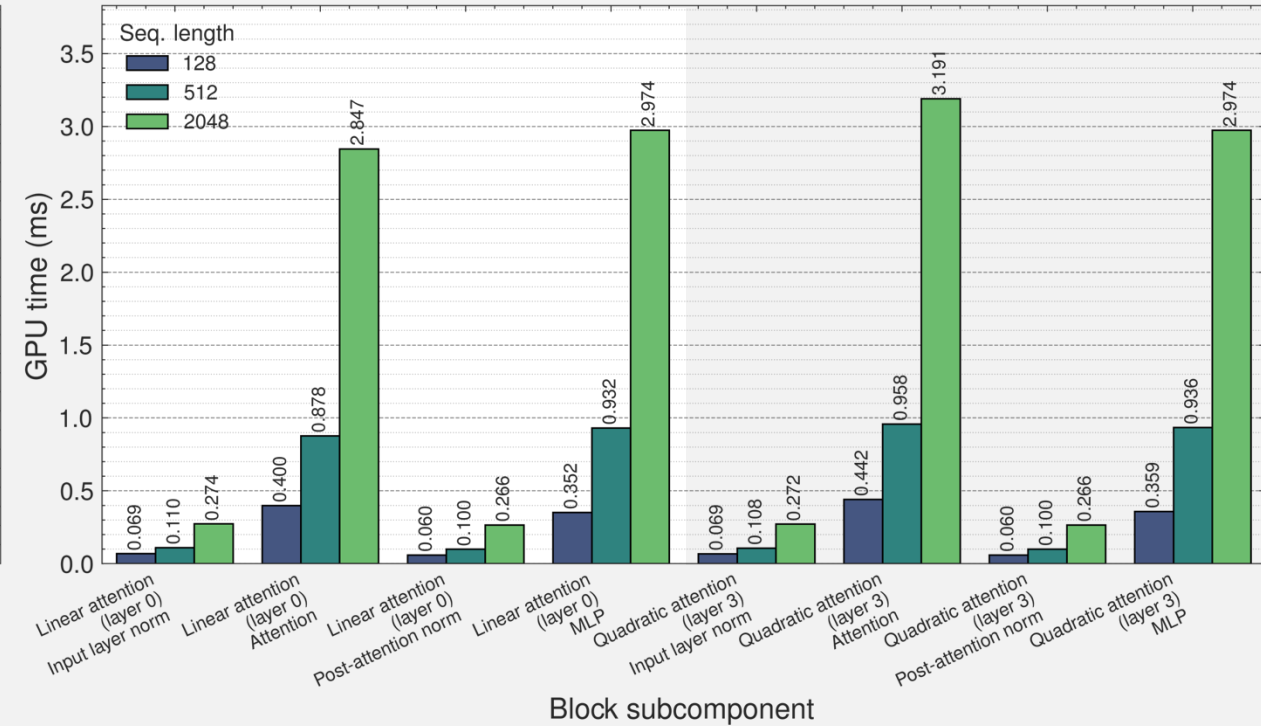
Qwen3.5-4B block / prefill

Qwen3.5-4B - prefill phase - bfloat16

Fixed sequence length: 2,048



Fixed batch size: 1



Input projection layer

- As first experiment I considered the quantization of both weights and activations of the input projection layer only

```
(62): FalconMambaBlock(
  (norm): FalconMambaRMSNorm(4096, eps=1e-05)
  (mixer): FalconMambaMixer(
    (conv1d): Conv1d(8192, 8192, kernel_size=(4,), stride=(1,), padding=(3,), groups=8192)
    (act): SiLUActivation()
    (in_proj): ActivationQuantizedSINQLinear(
      act_bits=8, granularity=per-token, name=backbone.layers.62.mixer.in_proj
      (linear): SINQLinear(in_features=4096, out_features=16384, bias=False, device=cuda:0, W_q.device=cuda:0)
    )
    (x_proj): Linear(in_features=8192, out_features=288, bias=False)
    (dt_proj): Linear(in_features=256, out_features=8192, bias=True)
    (out_proj): Linear(in_features=8192, out_features=4096, bias=False)
  )
)
```

Input projection layer

- Evaluated on Wikitext2
- For activations: symmetric RTN; for weights: SING

Model	bfloat16	in_proj @ W4A8		in_proj @ W4A4	
	Perplexity	Perplexity	KL divergence	Perplexity	KL divergence
Falcon-Mamba7b	5.204	5.344	0.043	282.89	4.010
Mamba2-2.7b	9.081	9.216	0.019	104897771	16.309

Next steps

- Derive latency improvements of the whole block
- Analysis of the activations distributions → apply transformations
- Quantize also the output projection layer